

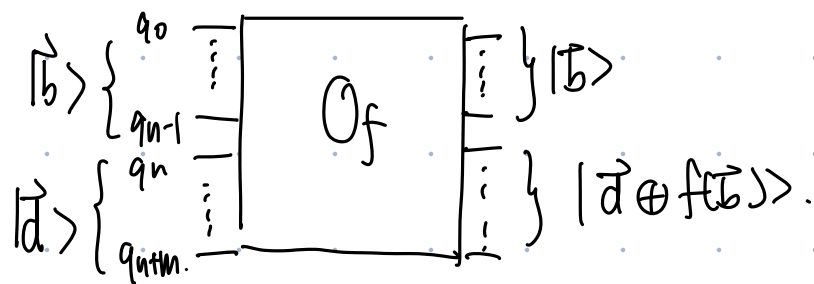
Lecture 17

- Quick Review of Deutsch Algo.

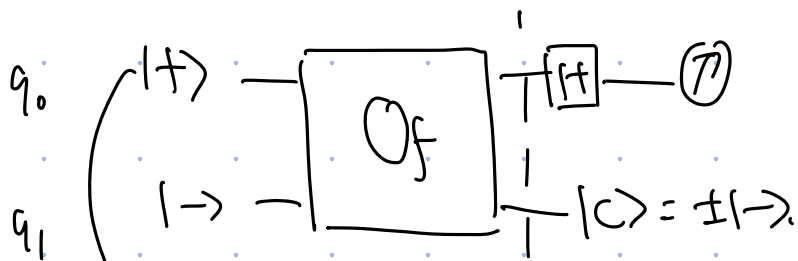
is $f: \{0,1\} \rightarrow \{0,1\}$ constant or balanced.

- Deutsch-Jozsa algo $f: \{0,1\}^n \rightarrow \{0,1\}$
exponential advantage

- Complexity $\Rightarrow$ Query Complexity.



$\rightarrow$ Deutsch Algorithm



ask 2 questions simultaneously : $f(0)$ & $f(1)$

$|\psi\rangle$

(q_1, q_0)

$$|\psi_1\rangle = 0 \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}} \otimes \frac{1}{\sqrt{2}} \sum_{x=0,1} |x\rangle \right\}$$

$$= \frac{|0 \oplus f(0)\rangle}{2} \otimes |0\rangle + \frac{|0 \oplus f(1)\rangle}{2} \otimes |1\rangle$$

$$- \frac{|1 \oplus f(0)\rangle}{2} \otimes |0\rangle - \frac{|1 \oplus f(1)\rangle}{2} \otimes |1\rangle$$

equivalently

$$= \sum_{x=0,1} \frac{1}{\sqrt{2}} \left\{ (|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle) \right\} \otimes \frac{1}{\sqrt{2}} |x\rangle$$

$$= \frac{1}{\sqrt{2}} \sum_{x=0,1} (-1)^{f(x)} |-\rangle \otimes |x\rangle$$

you can do $\oplus$

Handy identity

$$H|x\rangle = \frac{1}{\sqrt{2}} \sum_{z=0,1} (-1)^{x \cdot z} |z\rangle$$

$$|\psi_2\rangle = |-\rangle \otimes \frac{1}{2} \sum_{z=0,1} \sum_{x=0,1} (-1)^{f(x) + x \cdot z} |z\rangle$$

Define a_z = $\frac{1}{2} \sum_{x=0,1} (-1)^{f(x) + x \cdot z}$

amplitude

$$a_z \text{ of } z=0 = \frac{1}{2} \sum_{x=0,1} (-1)^{f(x)}$$

$= \pm 1 \rightarrow$ if f is constant.

if f is constant, measuring $|0\rangle$ is 100%.

$$a_z \text{ of } z=1 = \frac{1}{2} \sum_{x=0,1} (-1)^{f(x) + x}$$

f balanced.

P1: $f(0)=0$. you get $a_{z=1} = \frac{1}{2} [(-1)^0 + (-1)^1] = 0$.

P2: $f(0)=1$ you get $a_{z=1} = \frac{1}{2} [(-1)^1 + (-1)^1] = -1$

if f is balanced,
measuring $|1\rangle$ is 100%.

Deutsch - Jozsa:

$$f : \{0,1\}^n \rightarrow \{0,1\}$$

$$f(\vec{x})$$

$$Z(n,m) = 2^m 2^n = 2^{2n}$$

B.4
bit-strings
constitute a
vector space.

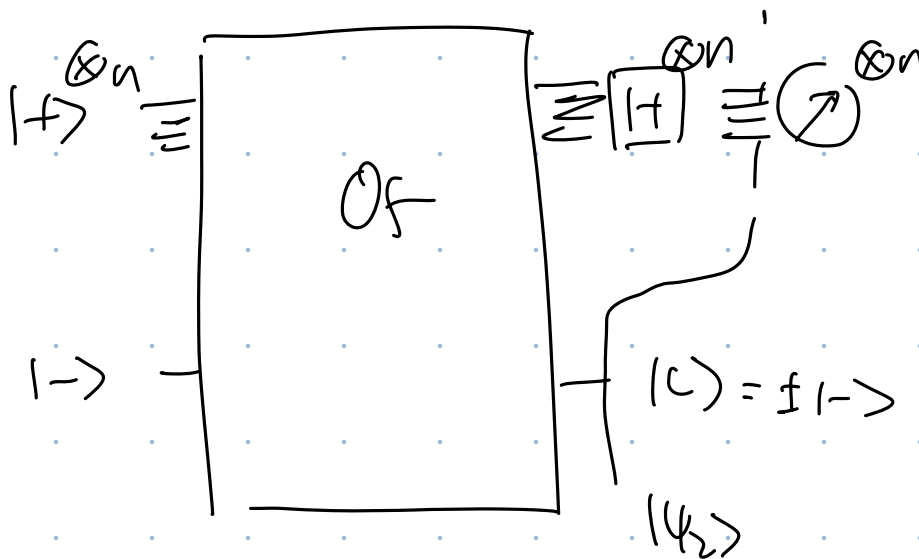
→ there are 2 of them that are constant → zero → one.

→ how many of them are balanced? interesting question to think about.

$$Is it \binom{n}{n/2} ? \quad \text{let } n=2m \quad \binom{2m}{m} \rightarrow \frac{(2m)!}{m! m!}$$

Alice creates oracle O_f .

Promise: f is constant or balanced.



→ Can you apply a control qubit to control qubit to

decide whether or not to apply the oracle.

→ what information does the global phase before the $\rightarrow$ tell you?

$$H^{\otimes n} |0\rangle = |+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{\vec{x}} |\vec{x}\rangle$$

$$|\psi_1\rangle = Of \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}} \otimes \frac{1}{\sqrt{2^n}} \sum_{\vec{x}} |\vec{x}\rangle \right\}$$

$$= \frac{1}{\sqrt{2}} \sum_{\vec{x}} \left(|0 \oplus f(\vec{x})\rangle - |1 \oplus f(\vec{x})\rangle \right) \otimes \frac{1}{\sqrt{2^n}} |\vec{x}\rangle$$

$$= \rightarrow \otimes \frac{1}{\sqrt{2^n}} \sum_{\vec{x}} (-1)^{f(\vec{x})} |\vec{x}\rangle$$

Analogy $H|z\rangle = \frac{1}{\sqrt{2}} \sum_{\vec{z} \in \{0,1\}} |\vec{z}\rangle \cdot (-1)^{\vec{x} \cdot \vec{z}}$

$$\Rightarrow H^{\otimes n} |\vec{x}\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{z}} (-1)^{\vec{x} \cdot \vec{z}} |\vec{z}\rangle \quad \vec{x} \cdot \vec{z} = \left(\sum_{j=0}^{2^n-1} x_j z_j \right) \bmod 2$$

$$|\psi_2\rangle = \rightarrow \otimes \frac{1}{2^n} \sum_{\vec{x}} \sum_{\vec{z}} (-1)^{f(\vec{x}) \oplus \vec{x} \cdot \vec{z}} |\vec{z}\rangle$$

For $\vec{z} = \vec{0} \rightarrow$ amplitude $a_{\vec{0}} = \frac{1}{2^n} \sum_{\vec{x}} (-1)^{f(\vec{x})}$ $\nearrow$ f const.
 $a_{\vec{0}} = \pm 1$
 $\rightarrow$ if f balanced $a_{\vec{0}} = 0$

→ one quantum query of Of .

Classical Complexity

- worst case scenario: $\frac{1}{2}2^n + 1 = 2^{n-1} + 1$ queries.

$$f(0000) = 1$$

$$f(0001) = 1$$

$\vdots$

$$f(0111) = 1$$

$$f(1000) = ?$$

- Probabilistic Algo

Choose inputs $\vec{x}$ randomly

if f is balanced, prob of

randomly getting $f(\vec{x})$ all the same in N queries

$P_M = 2 \times 2^{-M} \rightarrow$ probability of being fooled

$\begin{matrix} 0 & \text{or} & 1 \\ \swarrow & & \searrow \end{matrix} \rightarrow$ getting 0 each time $> \frac{1}{2}$